Related Roadmaps

- AI and Data Scientist Roadmap
- SQL Roadmap
- Python Roadmap
- Prompt Engineering Roadmap

Find the detailed version of this roadmap along with other similar roadmaps

roadmap.sh

Data Analyst

Building a Strong Foundation

Introduction

- What is Data Analytics
- Types of Data Analytics
  - Descriptive Analytics
  - Diagnostic Analytics
  - Predictive Analytics
  - Prescriptive Analytics

Key Concepts of Data

- Collection
- Cleanup
- Exploration
- Visualisation
- Statistical Analysis
- Machine Learning

Analysis / Reporting with Excel

- IF
- DATEDIF
- VLOOKUP / HLOOKUP
- REPLACE / SUBSTITUTE
- UPPER / LOWER / PROPER
- CONCAT
- TRIM
- AVERAGE
- COUNT
- SUM
- MIN / MAX

Learn Common Functions

- Charting
- Pivot Tables
- Learn SQL

Gain Programming Skills

- Python
- R

1. Learn a Programming Lang.
2. Data Manipulation Libraries
3. Data Visualisation Libraries

- Pandas
- Dplyr
- Matplotlib
- Ggplot2

Mastering Data Handling

Data Collection

- Databases
- CSV Files
- APIs
- Web Scraping

Data Cleanup

- Handling Missing Data
- Removing Duplicates
- Finding Outliers
- Data Transformation

Using Libraries for Cleanup

- Pandas
- Dplyr

Data Analysis Techniques

Descriptive Analysis

- Generating Statistics
- Visualizing Distributions

- Dispersion
  - Range
  - Variance
  - Standard Deviation
- Central Tendency
  - Mean
  - Median
  - Mode
  - Average
- Distribution Space
  - Skewness
  - Kurtosis

Data Visualisation

- Tools
- Libraries
- Tableau
- Matplotlib
- ggplot2
- Power BI
- Seaborn
- Charting
- Bar Charts
- Histograms
- Line Chart
- Stacked Charts
- Scatter Plot
- Heatmap
- Funnel Charts
- Pie Charts

Statistical Analysis

Learn to analyze relationships and make data driven decisions

- Hypothesis Testing
- Correlation Analysis
- Regression

Learn Different Techniques

Advanced Topics

Machine Learning

Machine Learning Types

- Reinforcement Learning
- Unsupervised Learning
- Supervised Learning

- Model Evaluation Techniques

Popular ML Algorithms

- Decision Trees
- Naive Byes
- KNN
- K-Means Clustering
- Logistic Regression

Deep Learning (Optional)

- CNNs
- RNN
- Neural Networks
- Learn the Basics

- Frameworks
  - Tensorflow
  - Pytorch
- Practice Training Models
- Image Recognition
- Natural Language Processing

Big Data Technologies

- Big Data Concepts
- Data Storage Solutions
- Data Processing Frameworks
  - Hadoop
  - Spark
- Data Processing Techniques
  - Parallel Processing
  - MPI
  - MapReduce

Build a portfolio of projects. Some example projects:

1) Predicting sales trends 2) Customer Segmentation.

Participate in Kaggle Competitions

Participate in competitions and learn from community

Online Courses and Certifications

Platforms like Coursera, edX, Udemy and Datacamp offer courses in Data Analytics and related topics.

Stay Updated and Network

Join networking events, communities, seek mentorship, attend webinars and engage in discussions to expand your network.

Keep learning with the following relevant track

AI and Data Scientist Roadmap